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Study of multicomponent sorptive gas flow in fractured porous media using in-situ synchrotron X-ray imaging technology

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Abstract

Carbon dioxide (CO₂) geo-sequestration in deep coal seams is a complex process, involving multicomponent gas diffusion, competitive gas sorption, and the associated mechanical deformation of the coal mass. There is no sufficient knowledge about these coupled underlying multi-physical behaviours, so an in-depth fundamental understanding of these processes is required. In this paper, we conducted in-situ core flooding experiments to study multicomponent gas flow dynamics in coal through microscale synchrotron X-ray imaging. Since xenon (Xe) and krypton (Kr) have high X-ray attenuation coefficients, they could be directly observed under X-ray imaging. Thus, we use Kr as analogues of methane (CH₄) and Xe to represent CO₂, due to similar sorption behaviours in coal. The high-resolution imaging results uncover the process of the multicomponent gas exchange and sorption, gas diffusion, and sorption-induced fracture deformation. We presented the direct evidence of competitive adsorption behaviours of different gas coal lithotypes types during the core flooding. The image data provide a method to quantify the mass transfer coefficients under different gas types and conditions, which further enable us to perform larger-scale gas Advection–Diffusion–Sorption modelling with lab-verified parameters. Moreover, we found that gas desorption by the mechanism of gas competition exchange is more dominant than sample depressurisation. When there are multiple sorptive gases, fracture deformation is not significant because gas adsorption and desorption have opposite effects on fracture aperture.

Keywords Geo-sequestration, Coal, Synchrotron micro-CT imaging, Fractured geological media, Permeability, Digital core analysis

1 Introduction

Storing carbon dioxide (CO₂) in underground geologic formations, known as CO₂ geo-sequestration or carbon geological storage, has been recognised as a practical and necessary pathway to achieve the global net-zero emission target, which is also a critical component of Carbon Capture, Utilisation, and Storage (CCUS) [1–5]. The deep unmineable coal seams, which can adsorb methane naturally, are considered as one of the target geological media with large capacity and sealing efficiency [6]. Since coal has a higher affinity to CO₂ than methane (CH₄), injected CO₂ will displace methane in the coal matrix. The gas adsorption/desorption process will induce coal matrix

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deformation, which in turn affect the CO₂ injectivity into the coal seams [7–9]. Geo-sequestration projects are not always successful because the coal matrix swells during CO₂ injection, leading to reduced injectivity of the coal seams, which has become a major challenge [10]. Although the bench studies of adsorption behaviours of CO₂ in coal have been learned well [11–15]. However, the process of CO₂ geo-sequestration involves complicated multiphysical phenomena, including multicomponent gas diffusion, competitive gas sorption on coal surfaces, gas reactions with the coal matrix, and associated swelling/shrinkage deformation of the coal mass [9]. These multi-physical phenomena are closely interrelated. One significant unresolved challenge is the inability to predict and characterise multicomponent gas diffusion and the resulting coal matrix deformation. Due to insufficient knowledge about above coupled underlying processes, an in-depth fundamental understanding about the multiphysical processes is required.

In recent years, advances in X-ray microcomputed tomography (micro-CT) imaging technology enable us to obtain microscale internal structure of geological materials [9, 16–27]. The resulting high-quality 3D digital images are valuable for modelling fluid flow and deriving petrophysical properties of rocks [28, 29]. Baatar, Mostaghimi [30] scanned the gas diffusion process in a coal sample using a benchtop micro-CT scanner, where the sample was connected with krypton gas and scanned every 24 h. Yuan, Jing [31] computed local diffusion coefficient based on a series of micro-CT images obtained during krypton gas diffusion within the sample. However, the benchtop micro-CT scanner requires several hours for scanning to achieve sufficient high resolution and image quality. This prolonged scanning time significantly limits our ability to study dynamic flow and adsorption behaviours, which have time-scale effects.

Synchrotron light source could increase the scanning speed significantly since it can emit a monochromatic X-ray beam with high photon flux without beam-hardening effect. Thus, the synchrotron-based X-ray imaging could achieve high spatial resolution down to micrometres but with much less acquisition time down to minutes or less. The fast-imaging technology has been applied for dynamic imaging of fluid flow and distributions in porous media [32–35]. It has also been used in the study of CO₂ geo-sequestration in coal. Zhang, Ranjith [36] studied the time-dependent evolution of 3D microstructures of coal after the injection of supercritical CO₂. Mayo, Josh [37] measured gas uptake over time in coal under ambient conditions using xenon (Xe), krypton (Kr), and CO₂ gases. However, a complete multiscale multiphysical modelling is yet to be validated against experimental results. Therefore, a more comprehensive experimental

design is required to better understand the complex multiphysical gas flow phenomena in coal.

In this work, we thus conduct in-situ synchrotron X-ray imaging of core flooding experiments for multicomponent gas dynamics in coal. This paper uncovers multicomponent gas exchange and diffusion within dynamically deforming coal structures. This work reveals the underlying multicomponent gas flow behaviour in coal, including gas exchange, gas diffusion in coal matrix, gas flow in coal fractures, and coal matrix deformation.

2 Methodology

2.1 Experimental setup and procedures

An X-ray CT transparent core holder made of polyetheretherketone (PEEK) is used in this experiment. PEEK is a suitable material for core holder as it has excellent mechanical performance and low X-ray attenuation coefficient [38]. The PEEK core holder is certified to stand 1500 psi and 98 °C. It is mounted on a rotation stage, to rotate (180°) the sample continuously while scanning. 1/16 in flexible PEEK tubing is used to connect the core holder to other components. The cylindrical coal sample is 6 mm in diameter and 12 mm in length. Pressurised gases from gas cylinders are injected into the confined sample and flow rates are measured at the outlet. Outlet pressure is maintained using a back-pressure regulator. A gas sampling port is installed at the end of the flow-line. For tests with the presence of multicomponent gas mixtures, gas samples are taken at different timesteps during the core flooding test. The gas sampling port has a septum for taking gas samples using a gas syringe and ball valves for controlling the gas flowrate. Gas composition and concentrations of collected gas samples are then measured using gas chromatography–mass spectrometry (GC–MS) analysis. Figure 1 shows the experimental setup.

In this work, four types of gases are used: carbon dioxide (CO₂), methane (CH₄), xenon (Xe), and krypton (Kr). Since Xe and Kr have high X-ray attenuation coefficients, they could be directly observed under X-ray imaging. We use Kr as analogues of CH₄, as they are found to have similar sorption behaviours in coal, and Xe is used to represent CO₂ as they have similar sorption-induced matrix deformation on coal [37, 39]. Table 1 shows four gases' swelling effects on coal and sorption capacity [39]. By comparing the resulting volumetric swelling of coal as a function of gas pressure, it was found that CO₂ and Xe had similar swelling effect on coal with 4.41 vol% and 4.34 vol% of the maximum swelling, respectively. On the other hand, CH₄ and Kr present similar swelling curves under increasing gas pressure, with 1.17 vol% and 1.33 vol% of the maximum swelling. In addition, Xe and Kr have similar sorption capacities on coal as CO₂ and CH₄.

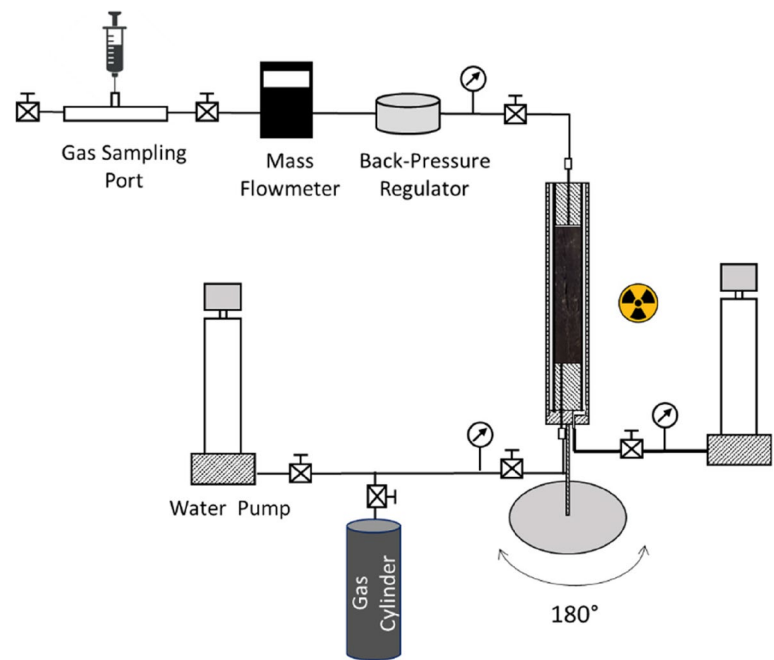


Fig. 1 Schematic experimental setup. X-ray CT transparent core holder is mounted on the rotation stage, its inlet is connected to the gas source and water pump. Confining pressure is applied on the sample via a hand hydraulic pump. At the outlet, gas outflow is sampled for gas concentration measurement

Table 1 Gas properties and tested swelling effect and sorption capacity of coal [39]

Properties	CO ₂	Xe	CH ₄	Kr
Maximum Swelling, vol%	4.41	4.34	1.17	1.33
Sorption Capacity, vol%	16.5	14.2	5.45	5.93
Molecular Weight, g/mol	44	131.3	16	83.8
Critical Temperature, K	304.2	289.7	190.5	209.4
Critical Pressure, MPa	7.38	5.84	4.6	5.5
Diameter, nm	0.33	0.40	0.38	0.36

Details of four core flooding tests are shown in Table 2. Test 1 investigates the adsorption process of Kr (analogue to CH₄), followed by CO₂ injection to displace Kr in Test 2. In Test 3, CH₄ is firstly injected into the sample under 74 psi of inlet pressure and 25 psi outlet pressure

to saturate the sample. Then, we inject Xe (analogue to CO₂) to displace adsorbed CH₄ under the identical gas pressures. In Test 4, the desorption process of Xe in sample is imaged while the sample is under vacuum. Test 3 and 2 mimic the CO₂ and CH₄ exchange process and sorption processes, where gas samples are collected at the outlet for further gas concentration measurements. All these dynamic processes are scanned.

2.2 Synchrotron X-ray CT imaging and processing

The synchrotron X-ray micro-CT imaging is performed at hutch 2B at Imaging and Medical beamline (IMBL) of Australian Synchrotron, ANSTO. The tomographic scans are acquired at different X-ray energy levels, depending on the K-edge of working fluids. In the case of Xe (K-edge: 34.55 keV) as the tracer gas, the scans are performed under 35 keV. In cases of Kr injections, the X-ray

Table 2 Fore core flooding tests details

Test	Working fluids	<i>P_{in}</i> , psi	<i>P_{out}</i> , psi	<i>P_{conf}</i> , psi	Gas sampling	Remark
Test 1	Kr	50	10	200	Yes	Inject Kr to dry sample
Test 2	CO ₂ , Kr	50	10	200	Yes	Inject CO ₂ to Kr-saturated sample
Test 3	Xe, CH ₄	74	25	200	Yes	Inject Xe to CH ₄ -saturated sample
Test 4	Xe	−14	−14	200	No	Vacuum Xe-saturated sample

energy levels are set to be 28 keV, which is the lower limit of the beamline energy and close to the adsorption edge of Kr (K-edge: 14.3 keV). The scanning parameters are listed in Table 3. The tomographic data is reconstructed using X-TRACT software installed on Australian Synchrotron's MASSIVE supercomputer cluster. Reconstructed image data are opened and visualised using ImageJ software. Prior to gas injections, the sample is scanned under dry condition, such that gas concentration can be determined by CT values difference between dry and wet sample.

2.3 Gas concentration measurement

We inject a 100 μ L gas-tight syringe into the gas sampling port to take gas samples. Collected gas samples are secured in 10 ml gas bags, which are taken to GC lab for composition measurement. Various premixed standards at volume fractions of 10%, 25%, 50%, 75%, 90% and 100% are used to obtain the calibration curve. The GC is an Agilent 7820 controlled by OpenLab CDS Chemstation Edition software. Split injection is used with a TCD detector, 3 ml/min. flow of Helium carrier gas through a short Molesieve capillary column 4 m $\times$ 320 μ m $\times$ 0.12 μ m. This reduced length enables reasonable peak shapes for carbon dioxide at elevated temperatures.

The inlet is set to 220 °C and 50 μ L of gas is injected. The other parameters depend on the gas being analysed. For xenon, the inlet split ratio is 80:1 and the oven temperature is 32 °C. For CO₂ and Kr the split is 120:1 and the oven temperature 170 °C. All gases with the exception of methane produced standard curves with linear regression coefficients, R^2 of at least 0.98. CH₄ is instead measured with an HP-5 column 30 m $\times$ 320 μ m $\times$ 0.2 μ m and FID detection. This provides a more reliable standard curve with an R^2 0.98. Kr elutes amongst the unretained oxygen and nitrogen peaks but still provide a viable standard curve.

This gas analysis is also checked with an Agilent 5975 C GC-MS with an DB-5MS UI capillary column, which does not separate the gases, but the extracted mass ions areas from Electron Ionization for each target gas are measured. Using a split ratio of 400:1 and injecting 5 or 10 μ L. This provides standard curves with R^2 of 0.97

or better, except for methane which is still problematic. There is a visible tapering of the peak area for all gases above 90%, which indicates the potential saturation of the EI detector. This at least confirms the reliability of the GC-TCD and the GC-FID results.

3 Results and discussions

3.1 Gas sorption behaviours

Figure 2 shows the Kr gas distribution during adsorption (Test 1) and desorption (Test 2) at selected different timesteps. The colour images represent the three-dimensional krypton adsorbing on dry coal (Fig. 2d-f) and desorbing under vacuum (Fig. 2j-l). The greyscale images are corresponding raw micro-CT images from the same cross-sectional area of the sample (Fig. 2a-c, g-i). We found that Kr gas molecules can widely invade into the coal matrix (Fig. 2a-f). Krypton gas can quickly occupy the inlet region of the sample in 8 min. Coal comprises of multiple lithotypes, including bright coal and dull coal [29]. Bright coal is rich in vitrain material and more brittle, which is permeated with fractures at right angles. Dull coal has dull lustre, composing of durian with fewer fractures developed [40]. It can be seen from Fig. 2a, there are 3 fractures observed in the bright coal in the selected slice, and one major fracture developed through dull and bright coal bands. We find that dull band is observed to adsorb more Kr gas than bright band (squared in Fig. 2c&f), despite of fewer fractures developed. The last scan of Test 1 is performed after Kr injection of 337 min, by which krypton has reached the whole length of the sample. However, there is a region without krypton adsorbed (the blue region in Fig. 2f). Based on the well-developed cleat network in that region, the blue region belongs to bright coal which has tighter coal matrix compared with dull coal. The overall gas concentration decreases from the inlet to outlet due to the inlet pressure (50 psi) is higher than outlet pressure (10 psi).

While injecting CO₂, Kr starts to quickly desorb from coal not only near the fractures, but also in the micropores of coal matrix (Fig. 2g-l). Since krypton is considered to be a close analogue of CH₄, where CH₄ gas molecule has a kinetic diameter of 0.38 nm, similar as Kr (kinetic diameter: 0.36 nm). Thus, Kr's sorption behaviour in coal gives significant information about CO₂ replacing CH₄ process in coal. CO₂ is more adherent to coal surface with much smaller molecule size (kinetic diameter: 0.33 nm) compared to krypton and CH₄, which means CO₂ should be much easier to spread within the coal micropores to replace adsorbed Kr/CH₄ molecules. In our previous work, we use [¹¹C]CO₂ as radiotracer to visualise CO₂ adsorption in coal using Positron Emission Tomography imaging [41], where we also found that CO₂

Table 3 Scanning parameters for different tracer gases in the synchrotron CT imaging

Working Fluid	Xe	Kr	CO ₂	CH ₄
X-ray Energy, keV	35	28	36.5	36.5
Exposure Time, s	0.25	0.5	0.25	0.25
Acquisition Time, min	7.54	15.08	7.54	7.54
Voxel Size, μ m	5.8	5.8	5.8	5.8

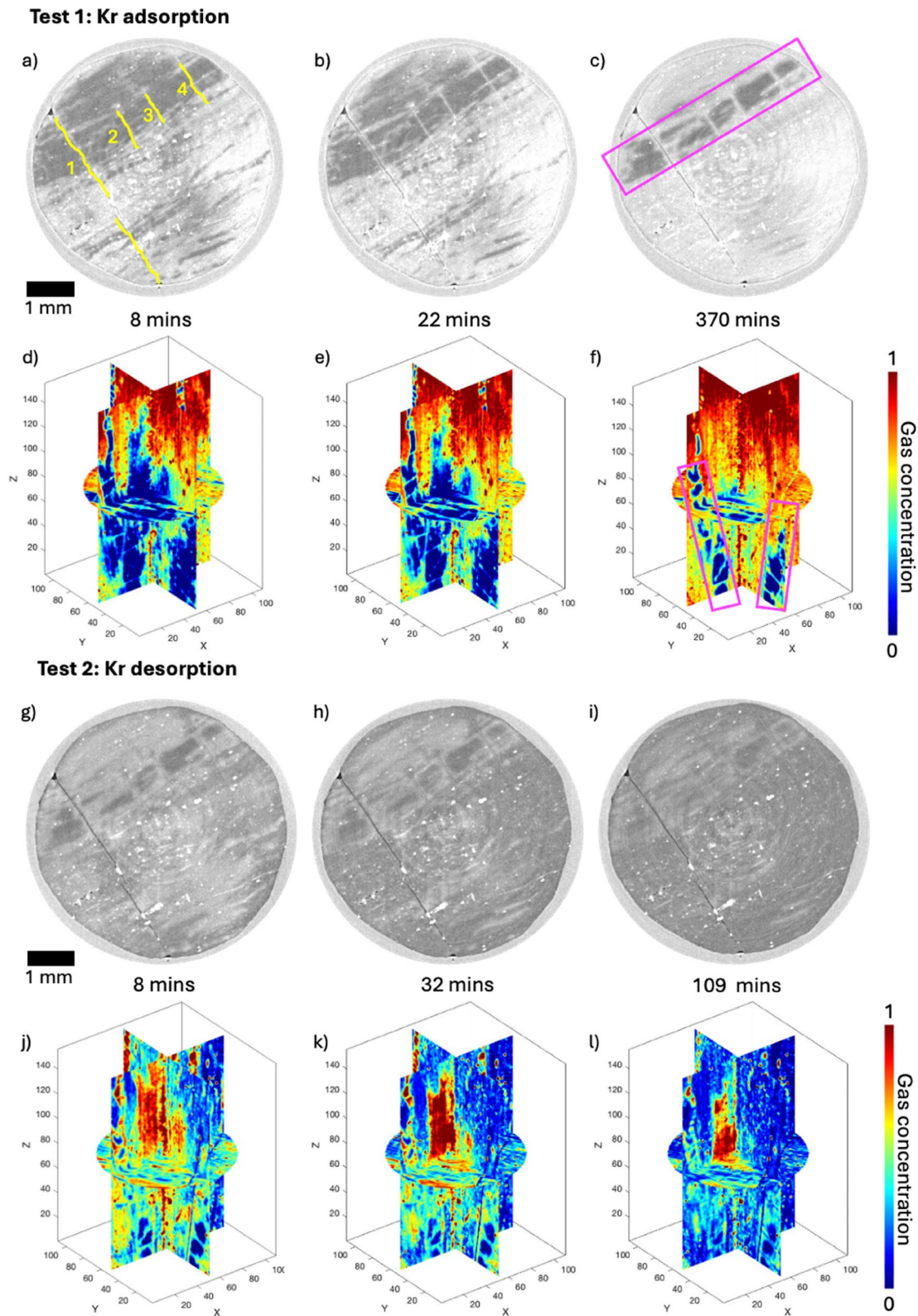


Fig. 2 Krypton gas distribution maps of at different timesteps during adsorption (Test 1) and desorption (Test 2). In the grayscale images, the lighter grayscale values represent higher gas concentrations. In the colour images, red to blue represent gas concentration from 1 to 0

can be adsorbed in micropores of coal matrix, not only near fractures.

However, we observe that Xe and CH₄ behave differently in coal compared with CO₂-Kr gas mixture (Fig. 3). Xe gas molecules are mostly adsorbed near the fractures even after 70 min, even though Xe has similar sorption capacity in coal with CO₂. It is because that the diffusion rate is largely determined by the kinetic diameter, with approximately four orders of magnitude reduction for each 0.1 nm increase in diameter [42]. Xe's molecule size (kinetic diameter: 0.40 nm) is larger than that of CO₂ (kinetic diameter: 0.33 nm). Thus, Xe penetrates the coal matrix significantly slower than to CO₂ to displace adsorbed CH₄, which is in agreement with Berens and Hopfenberg [42]'s findings. The molecule size not only determines the ease of gas transport in porous media but also affects Van der Waals force that bonds gas molecules on the coal surface. According to the Van der Waals force equation, Xe with larger gas molecule size has more electrons and greater surface area, leading to stronger Van der Waals force. In addition, the polarizability of Xe is also higher than that of CO₂, resulting in stronger temporary dipoles that cause greater attraction. This explains our observation of the desorption rate of Xe and CO₂, where Xe desorption is significantly slower. It is found that Xe is still visible in the sample even after 12-h continuous vacuum, due to stronger Van der Waals force and low diffusion rate. It emphasises the significance of investigating the gas-coal interactions for specific cases, since different gases tend to have significant different sorption behaviours within the coal sample.

For Test 2 and Test 3 with two gas components, gas samples at the outlet are collected at different scanning times to measure gas compositions using MS-GC (Fig. 4). In Test 3, Xe is injected into coal sample under the same pressure condition to displace methane. Injected Xe is not detected at the outlet at the beginning but accounts for 84% in the total gas outflow after 1 h. Together with our observation of Xe concentration map in Fig. 3, we conclude that most injected Xe flows out of the sample via fractures when the near-fracture coal matrix adsorbs enough Xe, rather than diffuse into deeper coal matrix for further adsorption. On the other hand, in the Test 2 of CO₂ replacing Kr, CO₂ concentration at the outlet only increase to 56% and stays a plateau until 200 min. It means nearly half of the injected CO₂ is adsorbed in the sample. This further illustrates the gas sorption behaviour differences between different gas mixtures.

3.2 Sorption location

Figure 5 shows the gas concentration change along the length of the sample at different timesteps. Gas concentration change is represented by the ratio of current CT

values and original CT values of the sample under dry condition. Regions with >100% mean gas adsorption, and regions with <100% mean that gas is desorbed there. Figure 5a is from Test 2 when Kr is displaced by injected CO₂, and Fig. 5b is obtained during Test 3 when Xe displaces methane and gets adsorbed. It is shown that, the increase of gas concentration is more significant at the inlet, which is closer to the Xe source. It means the methane component measured at the outlet is mainly desorbed from the sample inlet. However, the adsorption of Kr is more even through the whole sample length (Fig. 5a).

3.3 Sorption-induced deformation

Gas adsorption will cause coal matrix swelling, which in turn reduces fracture opening width. On the other hand, gas desorption leads to coal matrix shrinkage, enhancing coal permeability. To measure the permeability, gas flow rates and pressure differences are logged in Test 1–3 to determine permeability using Darcy's Law. Permeability values measured in different tests are shown in Table 4. It shows that partially CO₂-saturated coal sample has the lowest permeability as CO₂ has the most significant swelling impacts on coal compared with other gases. Partially Xe-saturated sample has the highest permeability, as Xe adsorption only occurs near the fractures according to Fig. 3.

To further study the impact of gas sorption on coal matrix deformation, we measure the fracture aperture sizes during gas sorption. Test 1 and Test 2 are scanned continuously under the same stress conditions, so that the mechanical effects on fracture opening is eliminated. We select a major fracture from the sample and extract the CT profiles in the direction perpendicular to its surface. CT profiles present two peaks at opposite fracture surfaces, based on which aperture deformation could be estimated by measuring the distance between the two peaks Fig. 6a. Details of this method can be found in our previous paper [18]. Figure 6b illustrates the normalised aperture change of the selected fracture from adsorbing Kr to adsorbing CO₂ while releasing Kr gas. It can be seen that fracture deformation is not obvious in Test 1 during Krpton adsorption. It is due to the aperture measuring method is limited by the micro-CT resolution, which means the accuracy is one voxel size. Thus, krypton adsorption-induced matrix swelling is not significant. When further injecting CO₂ to displace Kr, the fracture is quickly closing to 50% of its original opening width, which approves the strong swelling effect of CO₂ on coal matrix.

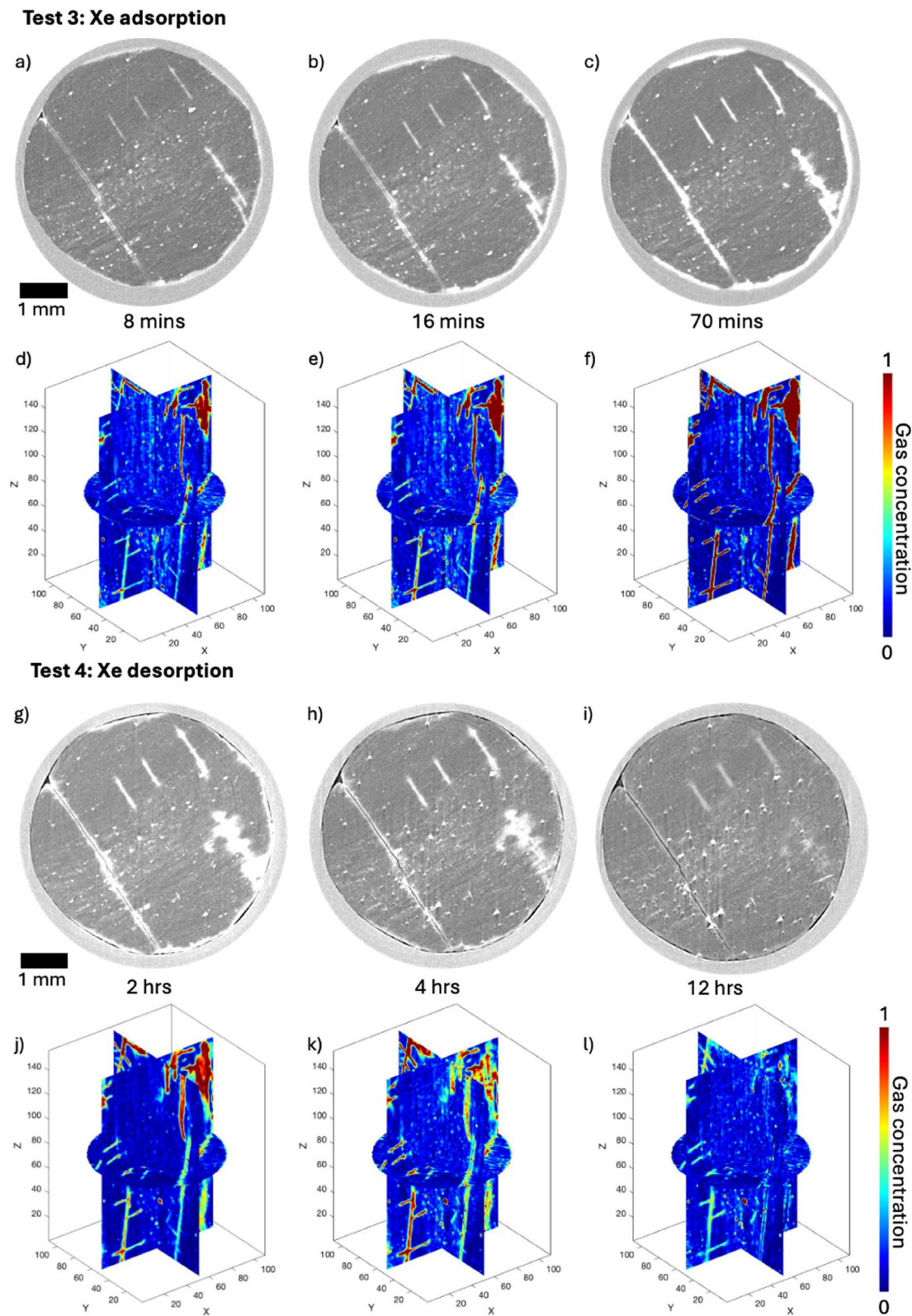


Fig. 3 Xenon gas distribution maps of at different timesteps during adsorption (Test 3) and desorption (Test 4). In the grayscale images, the lighter greyscale values represent higher gas concentrations. In the colour images, red to blue represents gas concentration from 1 to 0

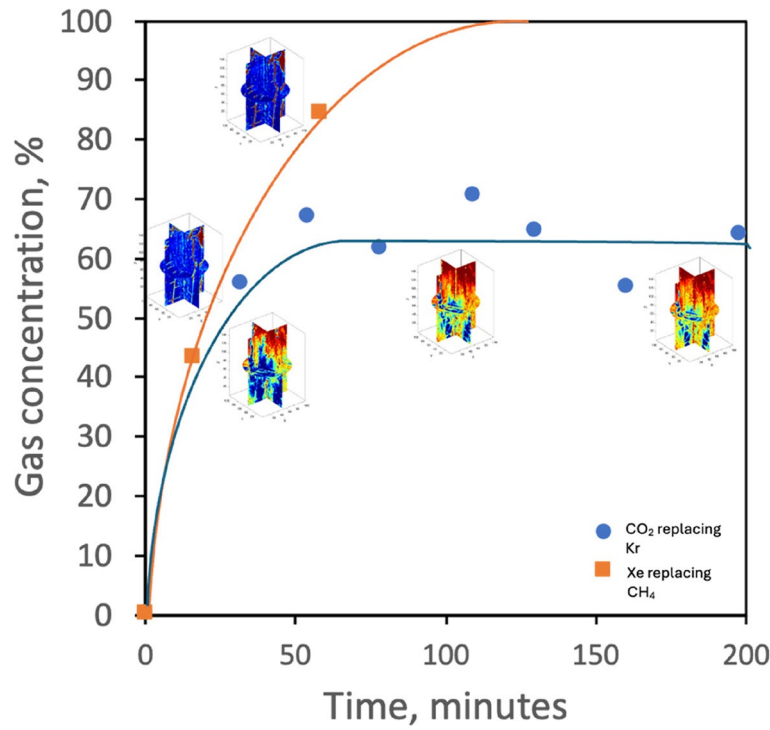


Fig. 4 Gas composition change during CO₂ replacing Kr (blue circle) and Xe replacing CH₄ (red square)

3.4 Gas sorption rate

By visualising the gas sorption processes, we found that different processes using different working gas types present different sorption rates. To quantify the sorption rates, linear driving force (LDF) model is used, which is proposed to describe adsorption chromatography [43]. It assumes that the mass transfer within the adsorbent solid is driven by the difference between the adsorbed phase amount $q_a(t)$ and solid/fluid equilibrium amount $q_a(\infty)$,

$$\frac{dq_a}{dt} = \beta[q_a(\infty) - q_a(t)]$$

where k is the overall mass transfer coefficient. Integrating the equation at from time 0 to t will give,

$$q_a(t) = q_a(0) + [q_a(\infty) - q_a(0)][1 - \exp(-\beta t)]$$

where $q_a(0) = 0$. From raw greyscale CT images obtained at different times, we can plot normalised gas concentration profile, $q_a(t)$. By fitting the CT data with LDF model, the over mass transfer coefficient β can be obtained (Fig. 7).

Table 5 lists the fitted mass transfer coefficient values for each test, the larger values indicate quicker sorption. Kr adsorption in dry coal sample has the largest mass transfer coefficient of 0.1025 min^{-1} , while its desorption rate is 19 times slower, which is because this adsorption is driven

by CO₂-Kr gas exchange rather than pressure difference. On the other hand, Xe gas is hardly released even under vacuum condition due to the gas molecules are bonded on coal surface by a stronger Van der Waals force under a lower diffusion rate, which has been discussion previously. In Mayo, Josh [37]'s work, they did not find the significant difference between Xe and Kr in terms of sorption rate, since they used different coal samples. But in this work, Xe sorption is slower than Kr as expected, even Xe is found to have higher sorption capacity than Kr in coal.

3.5 Gas concentration modelling

A larger cylindrical coal sample from the same coal mine with a diameter of 52 mm and the length of 60 mm is scanned using a HeliScan micro-CT scanner at the University of New South Wales [41]. The resulting resolution is $26 \mu\text{m}$. The sample domain is discretised into a $16 \times 16 \times 18$ model. To predict CO₂ adsorption in this coal sample under CH₄-saturated condition, Advection-Diffusion-Sorption Equation (ADS) is used,

$$\frac{\partial(\phi C_f)}{\partial t} = \nabla \cdot (\phi D_{eff} \nabla C_f) - \nabla \cdot (\mathbf{u} C_f) - \frac{\partial C_a}{\partial t}$$

where C_f is pore concentration of free gas, D_{eff} is effective diffusion coefficient, $\mathbf{u}$ is flow velocity, ϕ is porosity, and C_a is the adsorbed gas concentration per bulk volume, which is calculated using the LDF model:

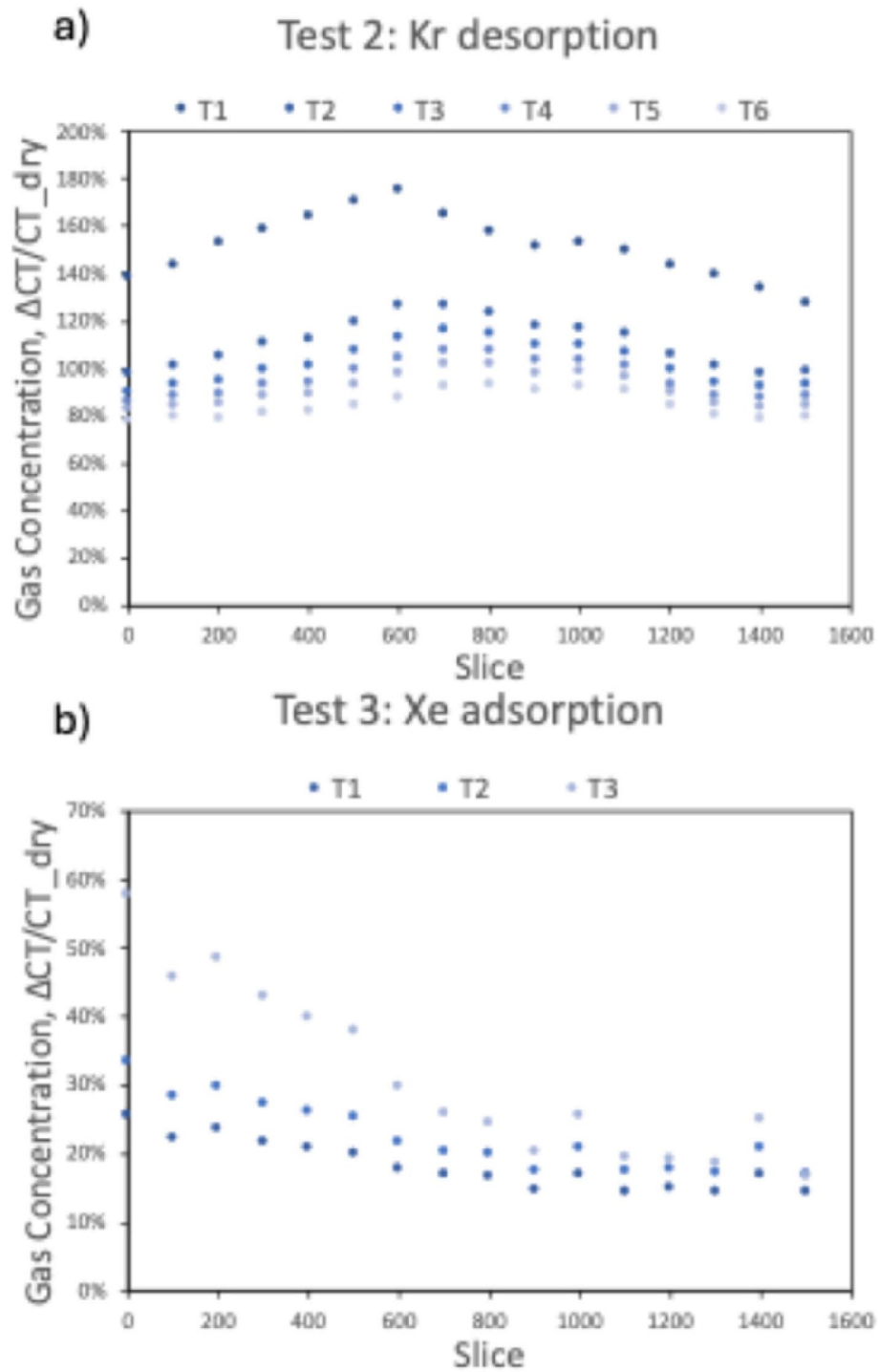


Fig. 5 Slice-averaged gas concentration along the length the sample at different timesteps. Darker blue indicates earlier timestep. Larger slice number means being closer to the inlet

$$\frac{\partial C_a}{\partial t} = \beta [C_a(\infty) - C_a(t)]$$

where the mass transfer coefficient β is obtained by fitting with CT images. In Test 2 of CO_2 displacing krypton,

we assume that the Kr concentration deduction is linearly correlated with CO_2 adsorption, thus $\beta = 0.020 \text{ min}^{-1}$ is used in this simulation. $C_a(\infty)$ follows a Langmuir isotherm,

Table 4 Measured permeability by core-flooding tests

Test	Sample Condition	Gas	Permeability
1	Kr saturated	Kr	90.06 mD
2	Partially CO ₂ saturated	CO ₂	56.41 mD
3	Partially Xe saturated	Xe	131.86 mD

$$C_a(\infty) = \frac{(K_{Lang}P)}{1 + K_{Lang}P}$$

K_{Lang} is Langmuir constant, P is gas pressure. Total gas concentration per bulk volume of coal is given by,

$$C = \phi C_f + C_a$$

MATLAB Reservoir Simulation Toolbox (MRST) is used to compute the velocity and gas pressure of each grid based on Darcy's Law equation, where porosity and permeability values are assigned to each grid block. Porosity of each grid block is obtained based on the segmented micro-CT images by calculating the ratio of fracture voxels and the total number of voxels,

$$\phi_{i,j,k} = \frac{\text{Num. of fracture voxels in grid } i,j,k}{\text{Num. of total voxels of each grid}}$$

Due to the limitation of micro-CT image resolution, the micro-porosity of the coal matrix is unable to be resolved, thus the porosity is solely fracture porosity. Therefore, permeability can be approximated using "cubic law" [44] based on porosity values,

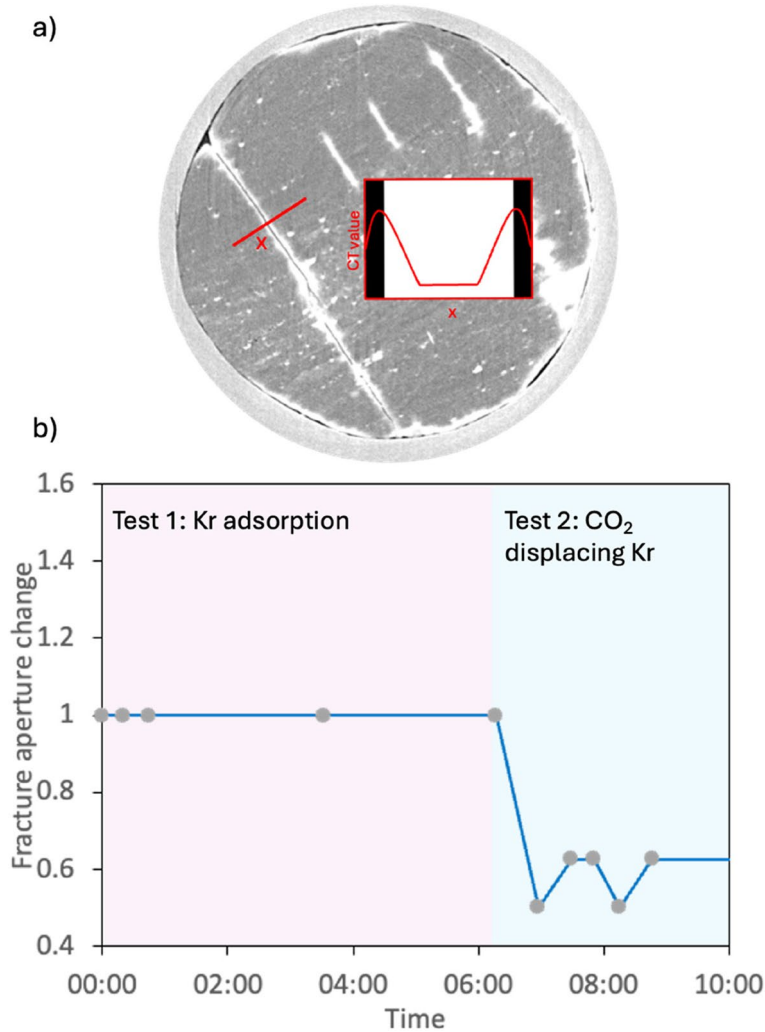


Fig. 6 The major fracture of the sample is selected for analysing aperture change (a). Its CT profile is extracted perpendicular to the fracture surface, which shows two peaks at two fracture surfaces. Aperture size is then estimated based on the CT profile. The aperture change at the same location of the same fracture is measured during Kr adsorption and desorption (b)

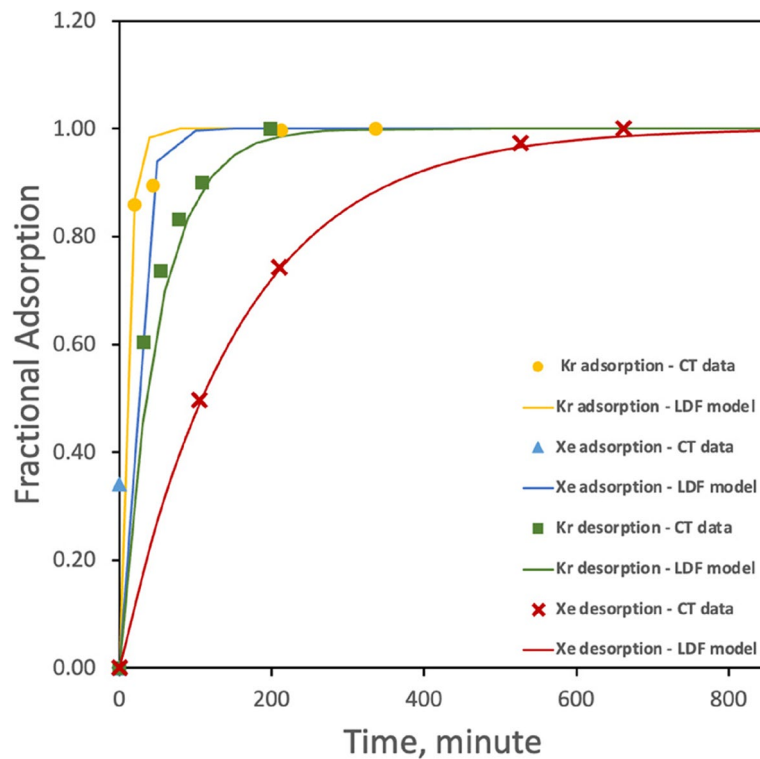


Fig. 7 Gas fraction adsorption curves measured from micro-CT images of four tests and fitted LDF curves

$$K_{i,j,k} = \frac{\phi_{i,j,k}^2 l}{12}$$

where l is side length of the grid block.

In this simulation, we model that CO_2 is injected from the top of the sample to displace adsorbed CH_4 . Figure 8 presents three-dimensional CO_2 concentration maps after 30, 60, 120 min of gas injection, where the total adsorption amount over 120 min is shown in Fig. 9. With the mass transfer coefficients obtained from the synchrotron data, gas advection–diffusion–sorption model for larger-scale CO_2 geo-sequestration simulation will be more certain. Further details of the ADS modelling can be found in our forthcoming paper.

4 Conclusions

During the process of CO_2 geo-sequestration in coal, there are multiple coupled physical phenomena happening, including multicomponent gas diffusion in micropores, competitive gas sorption on coal surfaces, gas reactive nature with the coal matrix, and associated swelling/shrinkage deformation of the coal mass. To have an in-depth fundamental understanding about the complex multiphysical gas flow phenomena in coal, we designed an experimental framework to dynamically observe multicomponent flow dynamics in coal with the application of synchrotron-based X-ray imaging. The image data provide a method to quantify the mass transfer coefficients under different gas types and conditions, uncovering multicomponent gas exchange and adsorption/desorption, gas diffusion, and sorption-induced fracture deformation. Both displacing and displaced gas components are imaged and studied under different injection scenarios. It further enables us to perform larger-scale gas Advection–Diffusion–Sorption (ADS) modelling with lab-verified parameters. In this work, we have the following findings:

- Different types of sorptive gas tend to have different behaviours in coal under the same conditions, since

Table 5 Fitted mass transfer coefficient of four tests

Test	Process	Mass Transfer Coefficient, β
1	Kr adsorption (dry sample)	$\beta = 0.1025 \text{ min}^{-1}, R^2 = 0.922$
2	Kr desorption (CO_2 displaces Kr)	$\beta = 0.020 \text{ min}^{-1}, R^2 = 0.994$
3	Xe adsorption (CH_4 saturated sample)	$\beta = 0.0562 \text{ min}^{-1}, R^2 = 0.995$
4	Xe desorption (under vacuum)	$\beta = 0.0064 \text{ min}^{-1}, R^2 = 0.999$

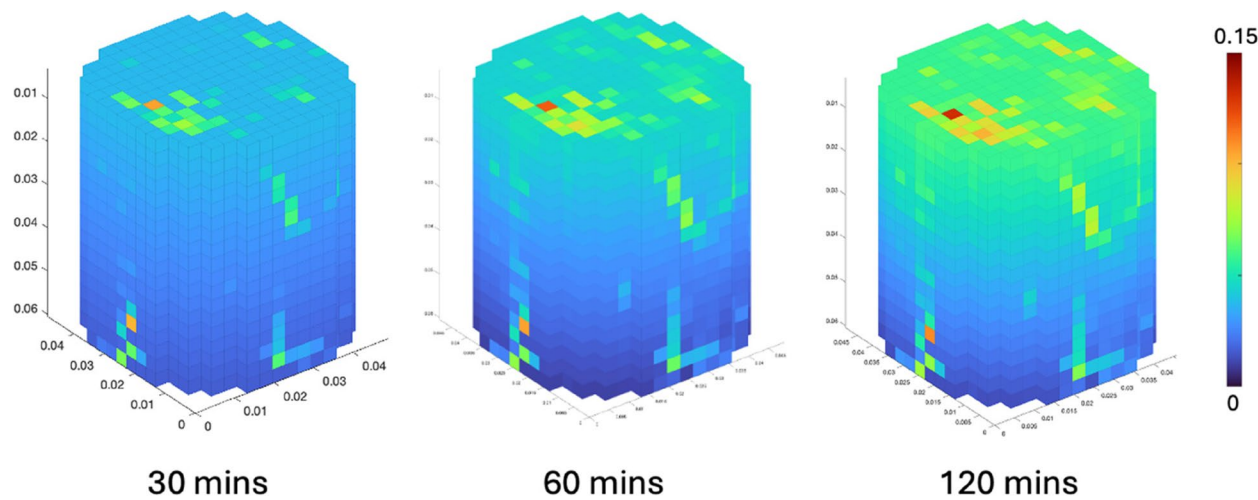


Fig. 8 Gas concentration map after 30 min, 60 min and 120 min

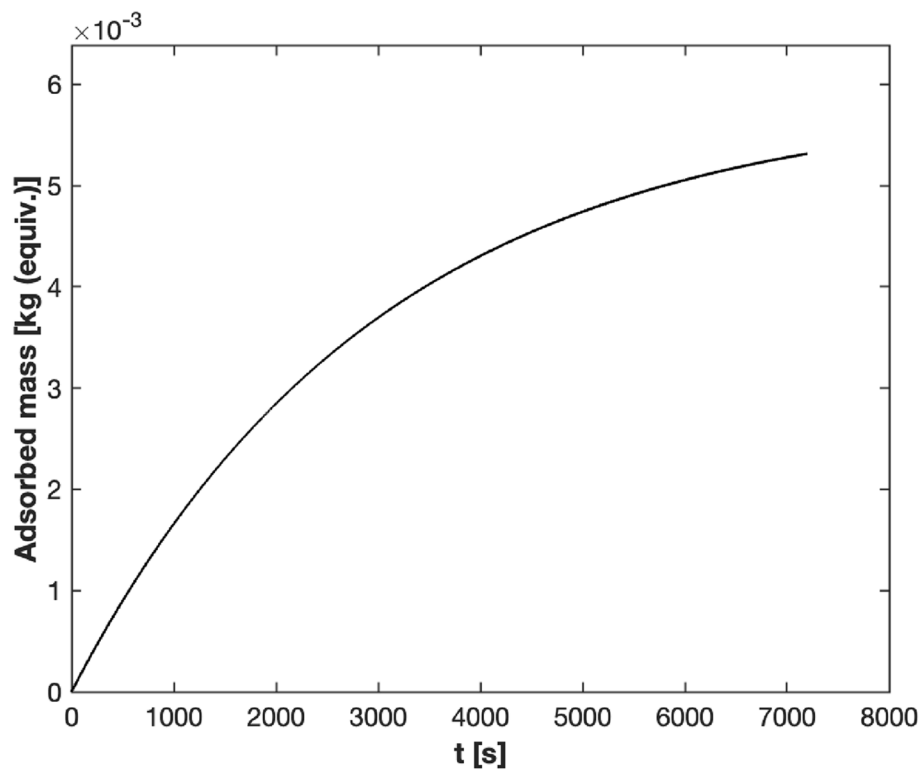


Fig. 9 Simulated CO₂ adsorption over time

the sorption behaviours are influenced by both diffusion rates and Van der Waals forces which are a function of gas molecule properties.

- Gas adsorption is also found to be dependent on coal lithotypes, where dull bands can adsorb more

Kr gas than bright bands, despite fewer fractures in those regions.

- Desorption and adsorption rates of the same gas are different. We find that Xe desorption is much slower compared to its adsorption rate, even given a pressure drop.

- Gas desorption by the mechanism of gas competition exchange is more dominant than depressurisation, especially when the adsorption is very strong (e.g. Xe). It indicates that CO₂ escape can hardly happen if there is no gas exchange.
- Fracture deformation is not significant for multi-component gas mixtures, where adsorption and desorption happen simultaneously with opposite effects on fracture aperture.
- Mass transfer coefficients of different gases can be obtained from the experimental image data, which can be further used for larger-scale ADS simulation to predict the gas concentration map over a longer timespan.

However, there are limitations in this work as well. Xe adsorption only happens near the fractures. To check if the gas can further diffuse into the coal matrix and get adsorbed like CO₂, a longer experimental duration is needed. Our future work will be combining this microscopic-scale image data with high temporal resolution (seconds) core-scale Positron Emission Tomography image data [41, 45] to predict a long-term, large-scale CO₂ plume migration in coal.

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Authors' contributions

YJ: Conceptualization, Data curation, Formal analysis, Funding acquisition, Methodology, Project administration, Supervision, Visualization, Funding acquisition, Writing – original draft. RA: Conceptualization, Data curation, Project administration, Writing – Review & Editing. MY: Data curation, Writing – Review & Editing. ZL: Data curation, Writing – Review & Editing. PH: Data curation, Writing – original draft. AUK: Data analysis, Modelling; PM: Conceptualization; Writing – Review & Editing; YZ: Conceptualization, Writing – Review & Editing; XZ: Supervision, Funding Acquisition, Writing – Review & Editing.

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Data availability

The data generated and analysed in this work are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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